



ONLINE COURSE:

**APPLICATION OF GIS
IN CONSERVATION
MAPPING**

**“Towards the Achievement of
Target 3 of the Global Biodiversity
Framework.”**

The banner features a background image of a gazelle on the left and a map of Eastern and Southern Africa on the right. The map shows various countries and water bodies, with some areas highlighted in green and blue. A small inset image of a cheetah is visible in the top right corner of the map area.

Erick Juma Wabwile
GIS Biodiversity Ecosystems
Expert



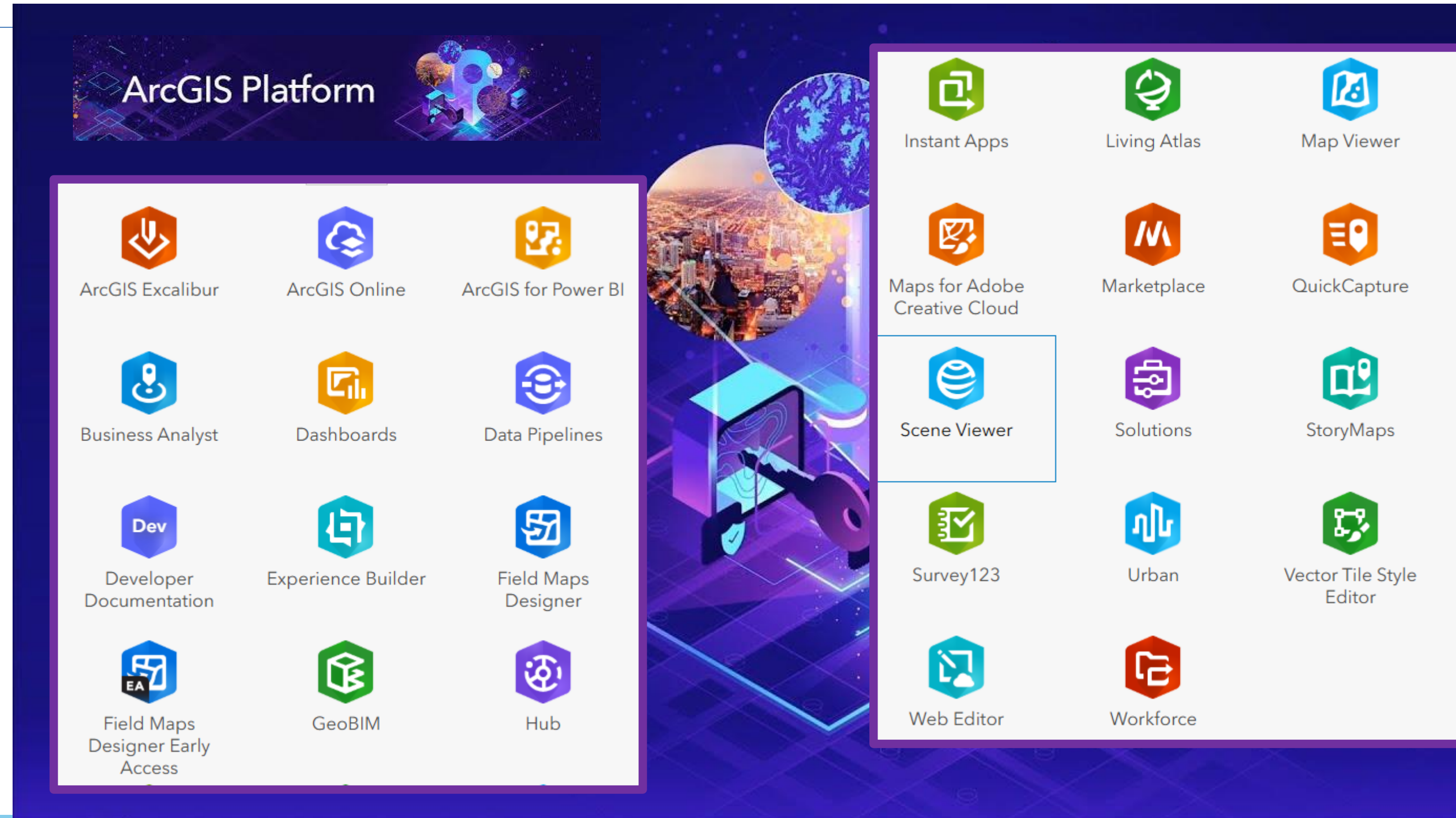
Module 1

Conservation Challenges & Global Biodiversity Framework

Session 3: GIS & RS

Overview of ArcGIS Platform

- ArcGIS is a geographic information system (GIS) platform developed by Esri. It's a comprehensive suite of **software, applications, and data** that allows users to collect, manage, analyze, and visualize geographic information.
- Essentially, it's a powerful tool for working with location-based data and creating maps, spatial analysis, and geographic data management.



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Creating a simple web-map

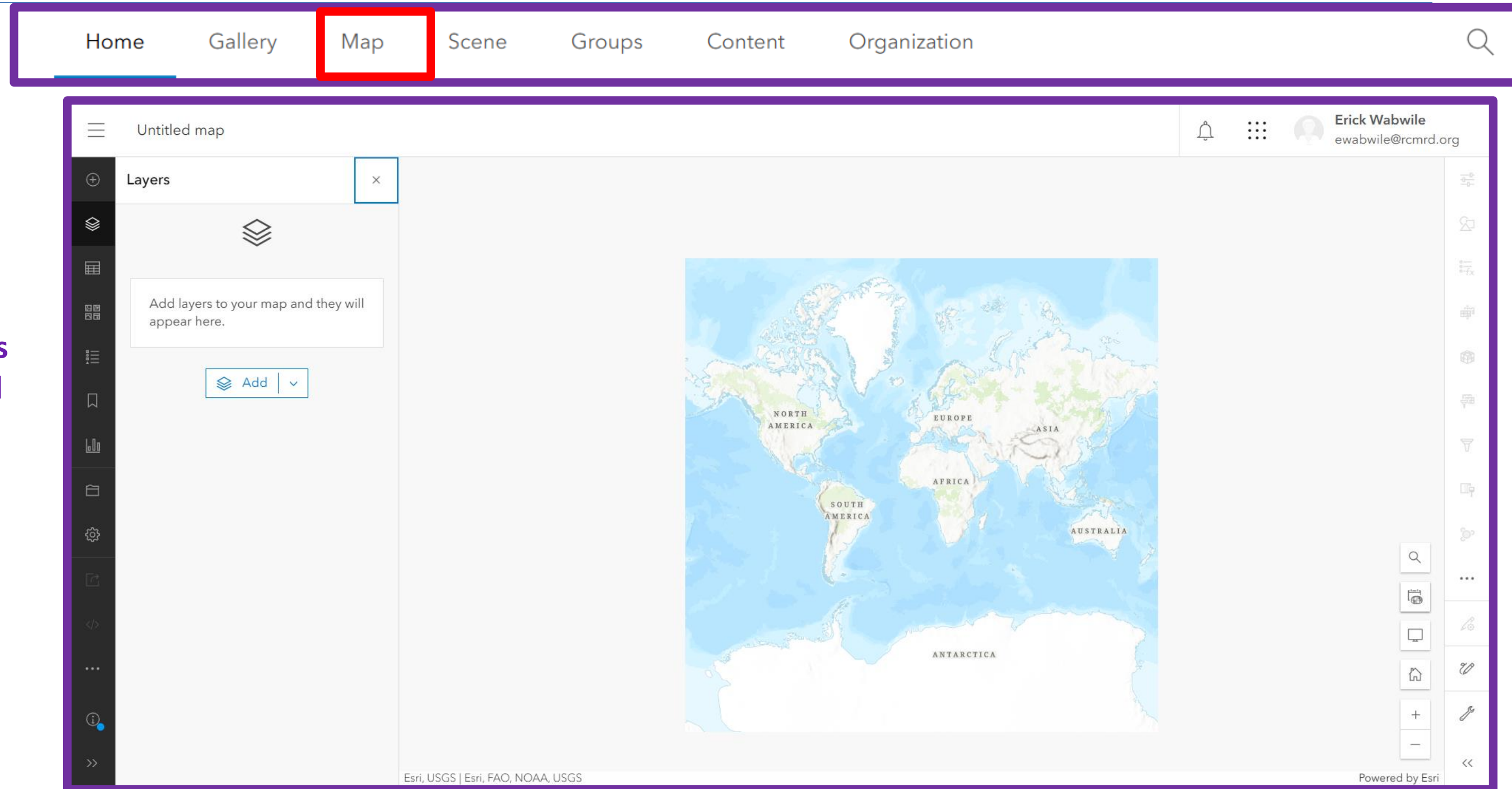
- Log into your ArcGIS Online Account. Please use the training account that was sent to you.
- You should be able to see the interface like the one shown on the right side of this slide

The screenshot displays the RCMRD ArcGIS Training Platform interface. At the top, a navigation bar includes links for Home, Gallery, Map, Scene, Groups, Content, and Organization, along with a search icon, a notification bell, and a user profile for Erick Wabwile (ewabwile@rcmrd.org). The main header features the RCMRD logo and the text "Welcome to RCMRD ArcGIS Training Platform". Below this, a large banner for the "Application of GIS in Conservation Mapping" course is shown, with the subtitle "Towards the Achievement of Target 3 of the Global Biodiversity Framework". The banner includes details: Duration: 3 Weeks (11 to 29 August 2025), Format: Online Training, Focus: 30x30 conservation target, and Certificate: Certified Course. Below the banner, four modules are listed, each with a circular icon and a ribbon indicating its level: Module 1 (Beginner Level) - Conservation Challenges & Global Biodiversity Framework; Module 2 (Intermediate Level) - Protected Areas Database Management; Module 3 (Advanced Level) - Mapping & Monitoring Habitat Loss; and Module 4 (Expert Level) - GIS for Wildlife Protection & Conservation. The footer of the banner reads "Conserve Nature. Empower Communities".

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Adding data to the Map

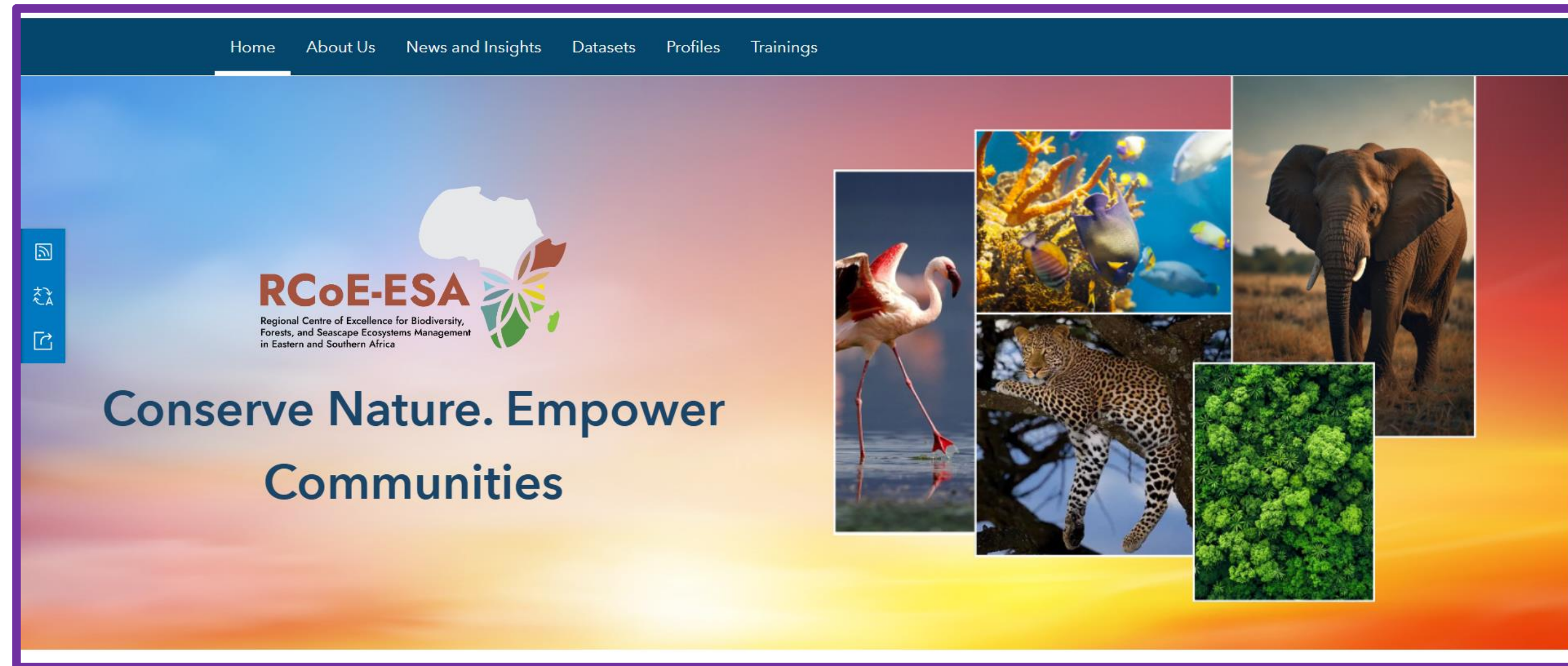
- Click on the map tab – you'll get an interface for creating a web map.
- Give your map a title. This time call it 'Protected and Conserved Areas of my country'



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Fetching data from various sources

- We will explore the various methods of adding data i.e., through api, uploading, and creating new data within the arcgis online platform.
- Visit the RCoE geoportal via this link <https://rcoe-geoportal.rcmrd.org/>



Fetching data from various sources

- [illegible]

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Fetching data from various sources

- Please click on **'I want to use this'** option.
- Click on the **'View API Resources'**
- Copy the **'Geoservice'** or the **'GeoJSON'** link then switch back to the ArcGIS online window

The screenshot displays the ArcGIS Online interface for the 'RCoE Protected Areas Polygons Data'. The main panel shows the dataset details, including the title 'RCoE Protected Areas Polygons Data', the publisher 'RCMRD Online', and the description 'Regional Centre for Mapping of Resource for Development'. Below this, there are buttons for 'View Full Details' and 'Download'. The 'Details' section lists the dataset type as 'Feature Layer', the update date as 'June 15, 2024 at 12:39:06 PM GMT+3', the data update date as 'June 15, 2024 at 9:39:06 AM GMT+3', the published date as 'June 15, 2024 at 9:24:14 AM GMT+3', the number of records as '5,055', and the access level as 'Public'. At the bottom of the main panel, the 'I want to use this' button is highlighted with a red box.

The 'View API Resources' panel is open, showing the 'GeoService' and 'GeoJSON' links. Both links are highlighted with red boxes. The 'GeoService' link is 'https://services6.arcgis.com/zOnyumh63cl' and the 'GeoJSON' link is 'https://services6.arcgis.com/zOnyumh63cl'. Below these links, there is a button 'Open in API Explorer' and a note 'Customize the query for your needs'. The background map shows the African continent with the Atlantic Ocean to the west and the Indian Ocean to the east.

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Fetching data from various sources

- Click on the Add icon  and choose 'Add layer from URL' option.
- Paste the link you copied under the URL section.

Untitled map

  Browse layers

  Add layer from URL

  Add layer from file

  Create sketch layer

  Create route layer

  Add media layer



Add Layer

URL

https://services6.arcgis.com/zOnyumh63cMmLBBH/arcgis/rest/services/RCoE_Protected_Areas_Polygons_Data/FeatureServer/0/query?outFields=*&where=1...

Supported layer types are CSV, GeoJSON, OGC WFS web service, OGC WMS web service, OGC WCS web service, OGC WMTS web service, OGC API - Features, ArcGIS Server web service, KML, GeoRSS, tile layer

Type

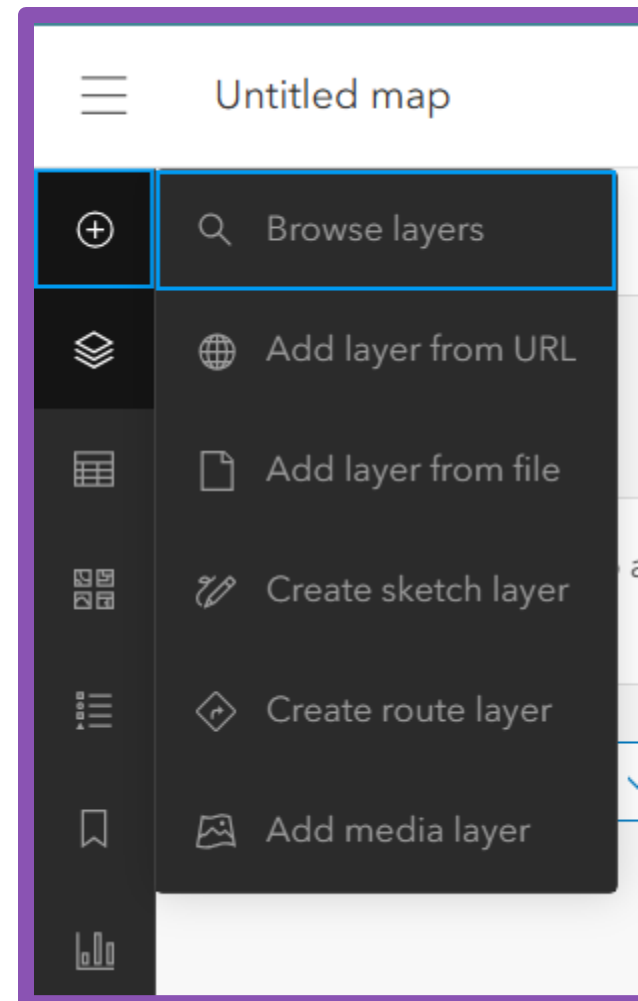
Select layer type

Cancel Add to map

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Fetching data from various sources

- Click on the Add icon  and choose 'Add layer from URL' option.
- Paste the link you copied under the URL section.
- The type of data will appear below, if you copied the GeoJSON link it will show there
- Click on the 'Next' button to proceed



Add Layer

URL

https://services6.arcgis.com/zOnyumh63cMmLBBH/arcgis/rest/services/RCoE_Protected_Areas_Polygons_Data/FeatureServer/0/query?outFields=*&where=1%3D

Supported layer types are CSV, GeoJSON, OGC WFS web service, OGC WMS web service, OGC WCS web service, OGC WMTS web service, OGC API - Features, ArcGIS Server web service, KML, GeoRSS, tile layer

Type



GeoJSON

An open standard geospatial data interchange format that represents simple geographic features.

Cancel

Next

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- Select 'Reference the file from the URL and add it to the map'. *No need to create a hosted feature layer at this point.*
- Then click on 'Add to map' button.
- You should be able to see the screen like the one shown on the right screen.

Fetching data from various sources

Add Layer

URL

https://services6.arcgis.com/zOnyumh63cMmLBBH/arcgis/rest/services/RCoE_Protected_Areas_Polygons_Data/FeatureServer/0/query?outFields=*&where=1%3D1&f=geojson

How would you like to add this file?

☐ Create a hosted feature layer and add it to the map

Utilize all of the capabilities a feature layer provides, including editing. Manage data completely within ArcGIS Online.

☒ Reference the file from the URL and add it to the map

Add data without creating a hosted feature layer. Ideal for smaller datasets that don't require features like editing. The data is loaded every time the map is opened.

Untitled map

Erick Wabwile
ewabwile@rcmr.org

Layers

RCoE Protected Areas
Polygons Data

Add

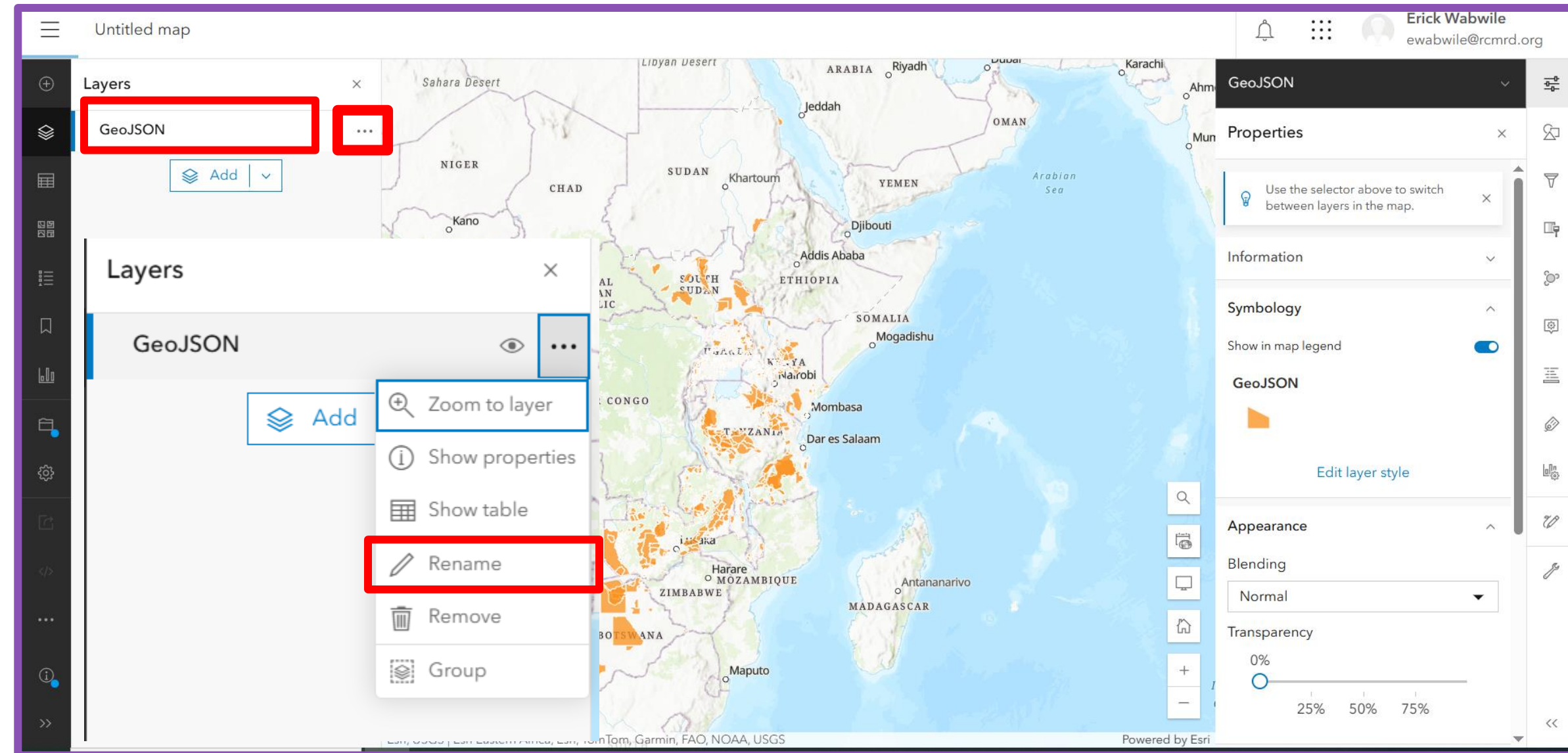
Esri, USGS | Esri, TomTom, Garmin, FAO, NOAA, USGS

Powered by Esri

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Renaming Layers

- If you used the GeoJSON link to add data, it may not capture the layer name.
- Click on the three dots beside the layer name and select 'Rename' option
- Change the name GeoJSON to 'Protected and Conserved Areas in Eastern and Southern Africa'



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Fetching data from various sources

- Repeat the process and add the protected areas points data under the RCoE geoportal datasets.
- This time round click on the 'View Data Source' option and copy the link in the url section at the top of the page.
- Then switch back to the ArcGIS online tab and add layer via the url option

The screenshot displays the ArcGIS REST Services Directory interface. The browser's address bar shows the URL: `services6.arcgis.com/zOnyumh63cMmLBBH/arcgis/rest/services/RCoE_Protected_Areas_Polygons_Data/FeatureServer/0`. The page title is "ArcGIS REST Services Directory". The breadcrumb navigation is "Home > services > RCoE_Protected_Areas_Polygons_Data (FeatureServer) > RCoE_Protected_Areas_Polygons_Data". The layer name is "Layer: RCoE_Protected_Areas_Polygons_Data (ID:0)". The "View In:" section shows "Map Viewer". The "Name:" field contains "RCoE_Protected_Areas_Polygons_Data".

On the left, there are five action buttons:

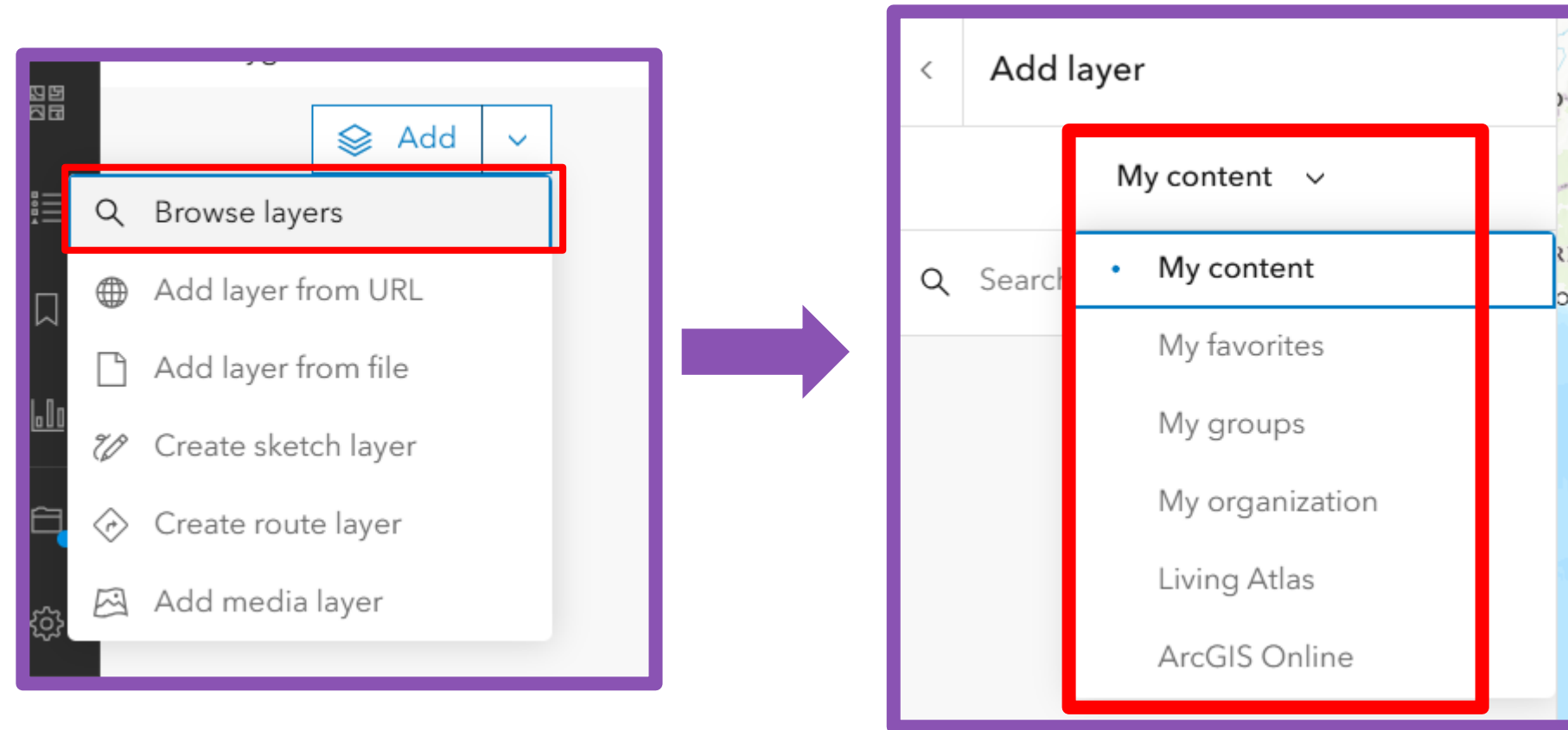
- Create a Map**: Start a map with this data
- Create a Story**: Open in ArcGIS StoryMaps
- View API Resources**: Try out the API Explorer
- View Data Source**: Select to open in a new tab (highlighted with a red box)
- Open in ArcGIS Online**: Select to open in a new tab

On the right, the "Add Layer" dialog box is open. The "URL" field contains the same URL as the browser address bar. The "Type" dropdown menu is set to "Select layer type". The "Add to map" button is highlighted with a blue box.

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Fetching data from various sources

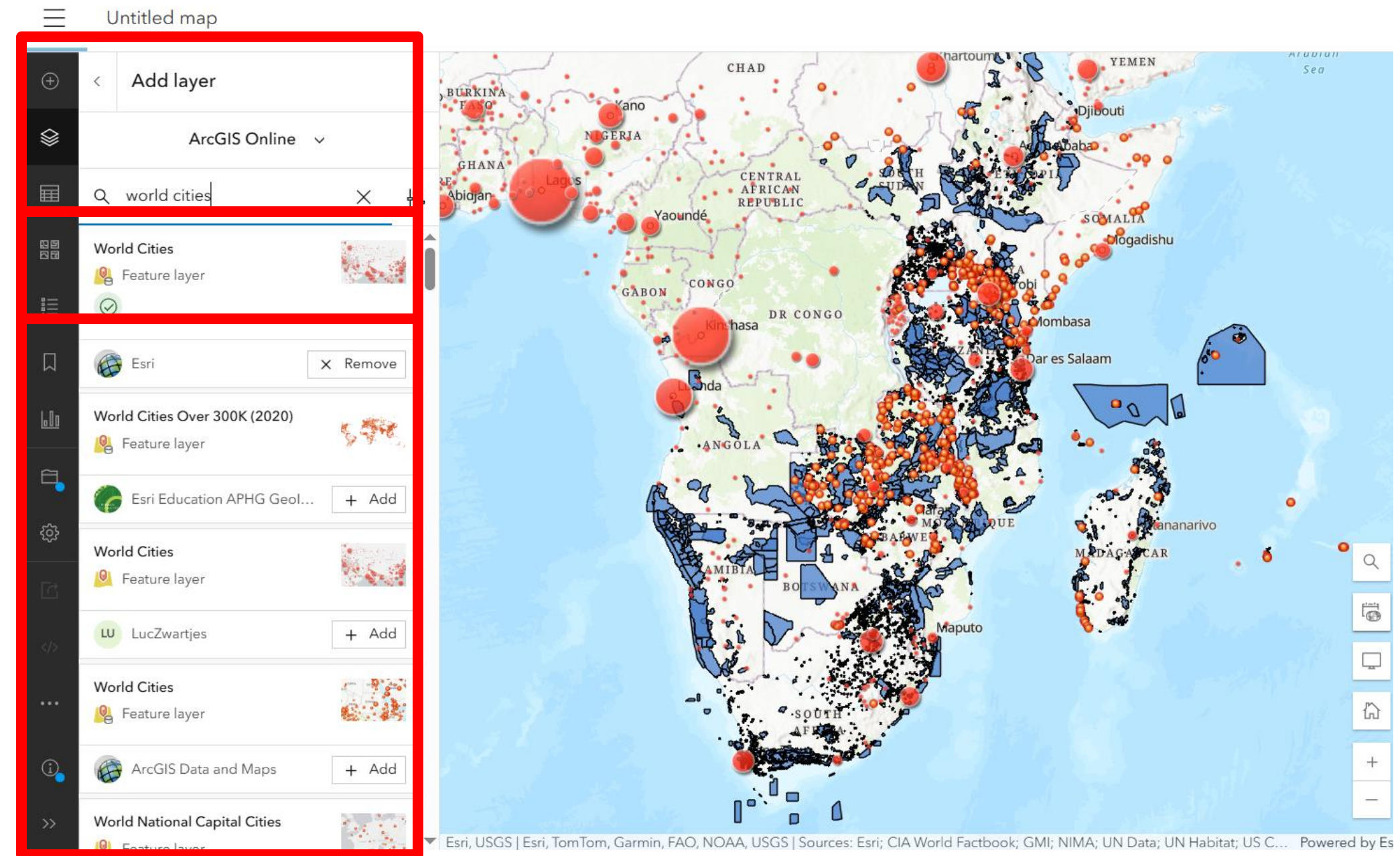
- Please note that you can also browse and add layers from your content, your favorites, your groups, your organization, living atlas, and shared content in ArcGIS Online



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Fetching data from various sources

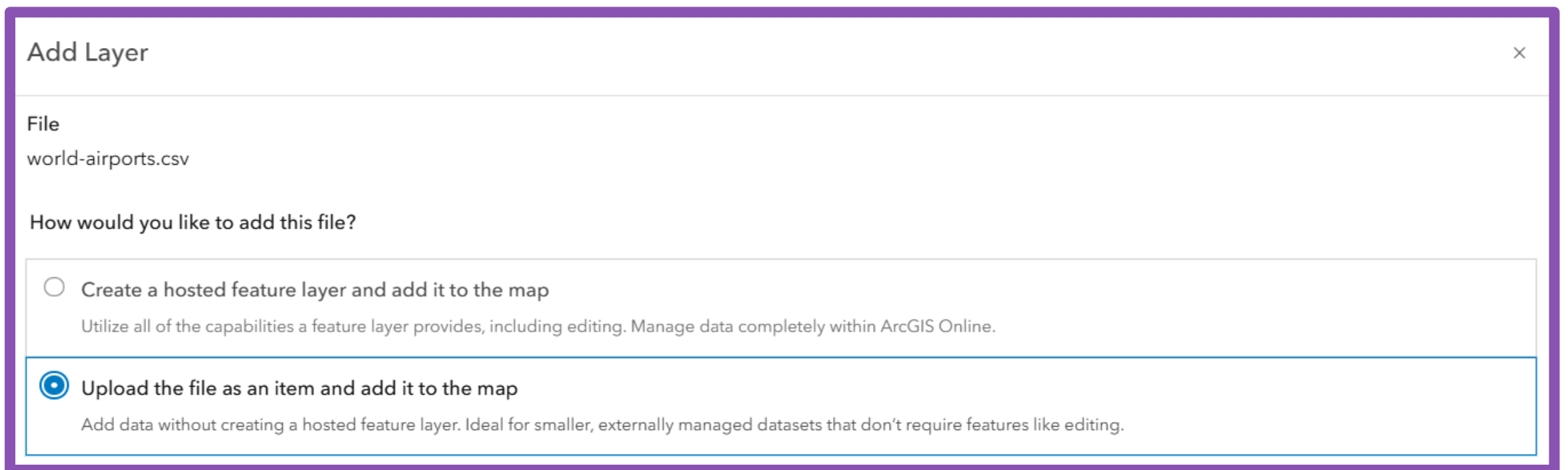
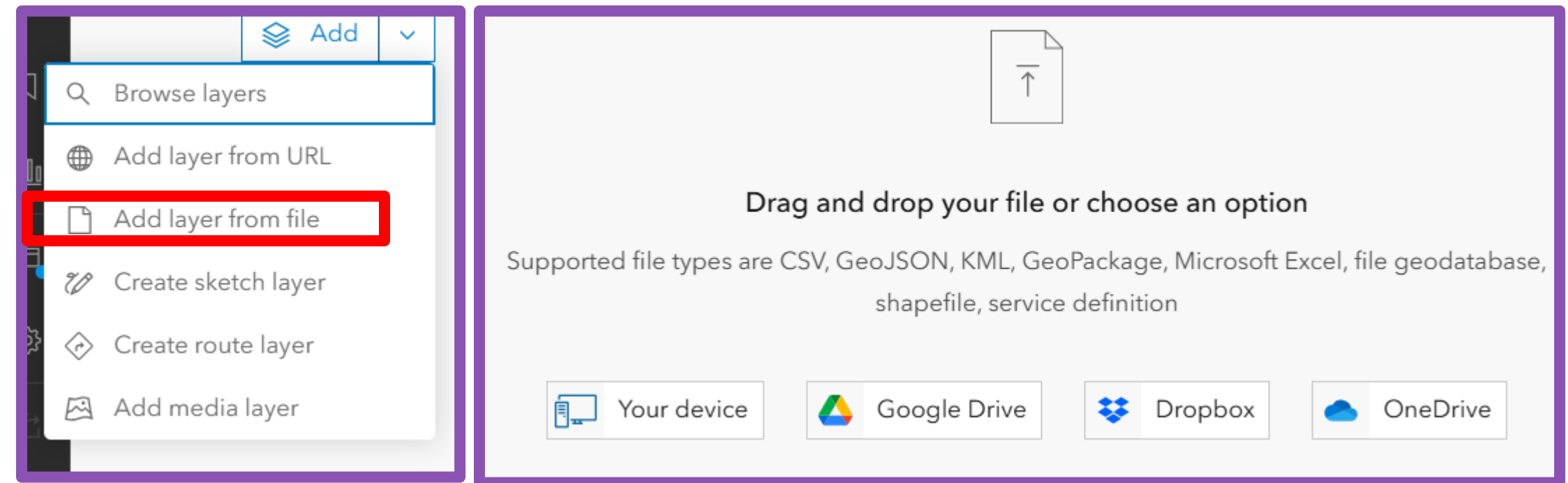
- Select ArcGIS online option and search for the Authoritative ☒ World Cities layer and add it to the map.
- Your map should look like the one on the right.



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Adding data from a file

- You can also add data from a file located on your computer or Google Drive, or Dropbox or OneDrive. Formats of a file that can be added include: shapefile, geodatabase, csv file e.t.c.
- To add data click on 'Add' and choose 'Add layer from file' option.
- Download the world international airports csv file via <https://ourairports.com/airports.csv> and add it to the map via the 'Add data from a file' option.



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Adding data from a file

- You can change the way the fields will be displayed and specify the Latitude and Longitude fields. And then click on the ‘Next’ button.

Add Layer

Location settings

Specify the type of location information the file contains.

 Latitude and longitude

Location data is latitude and longitude.

Location fields *

Location type	Field
Latitude	latitude_deg
Longitude	longitude_deg

Back

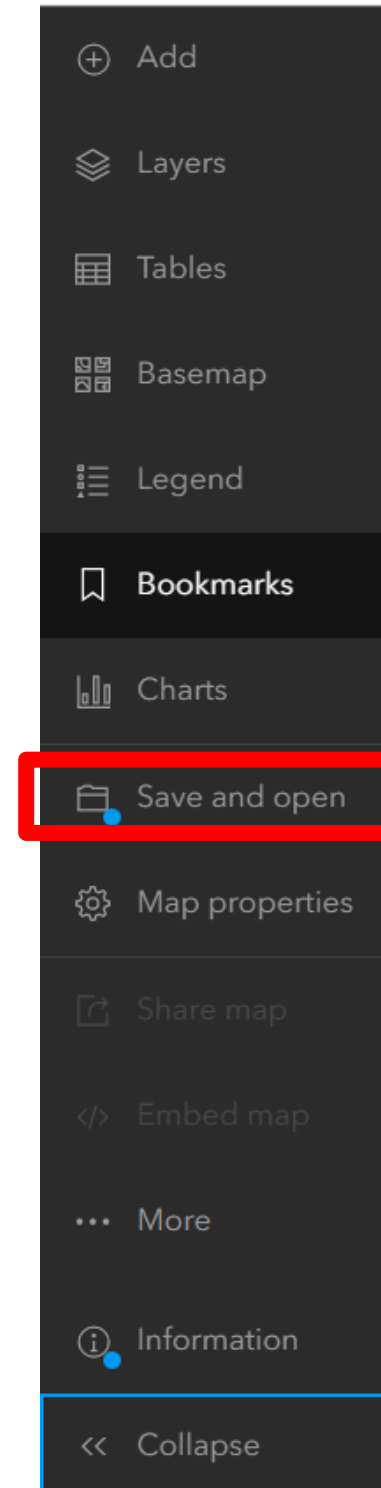
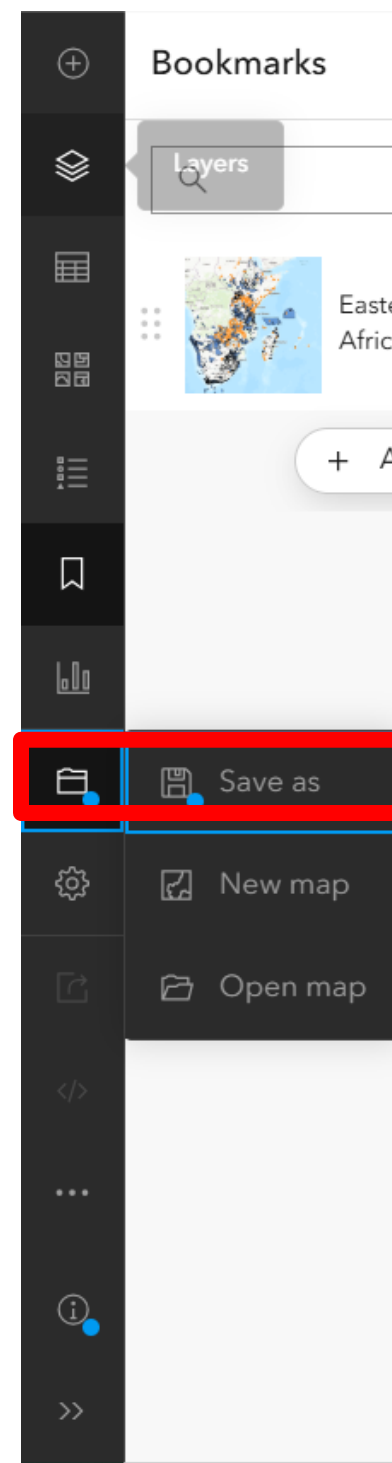
Cancel

Next

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Saving the map

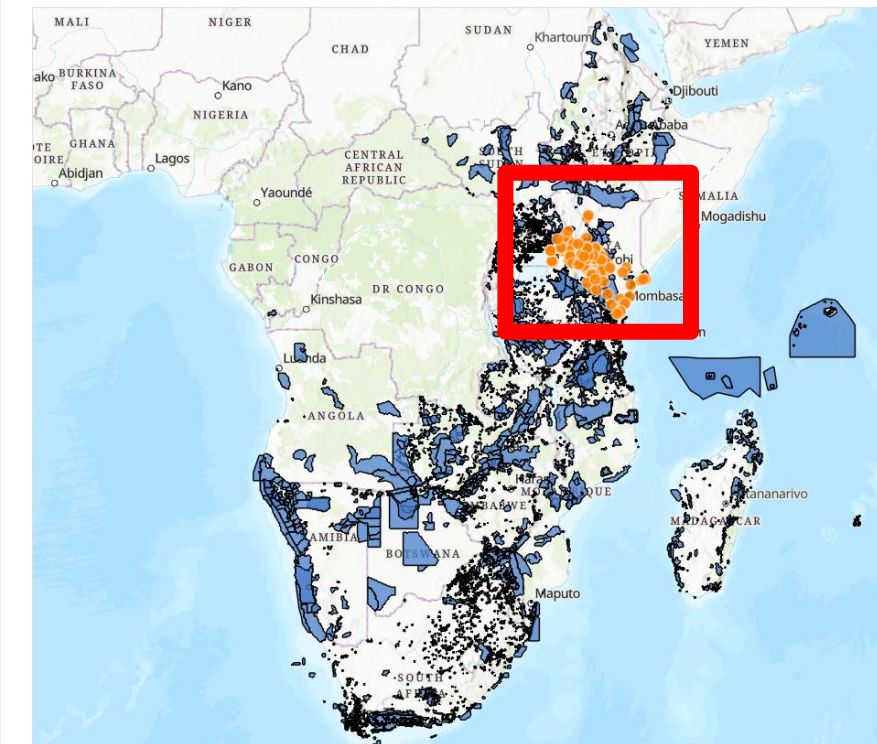
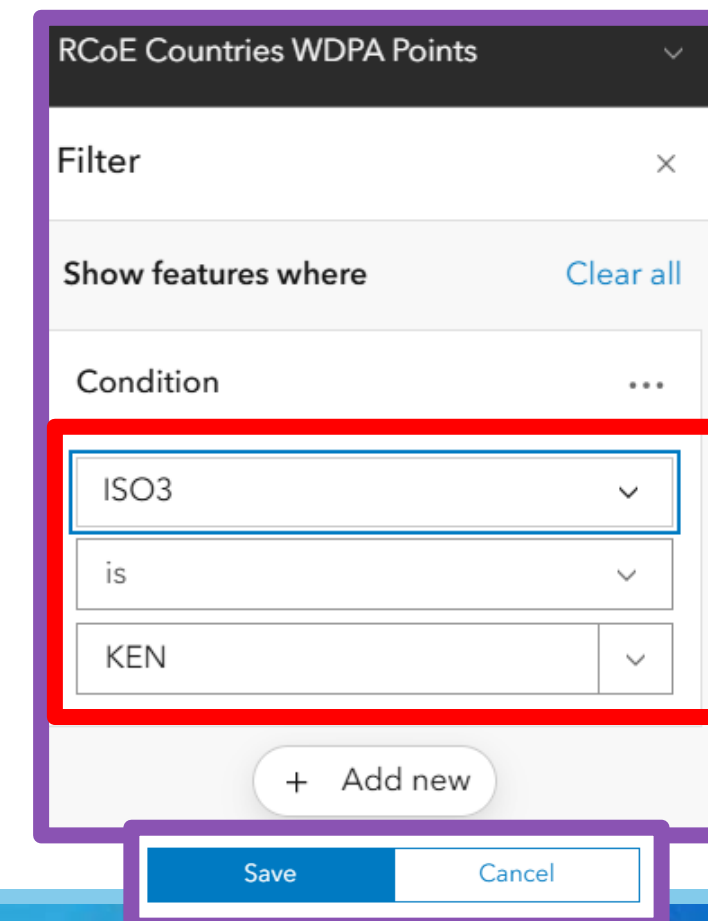
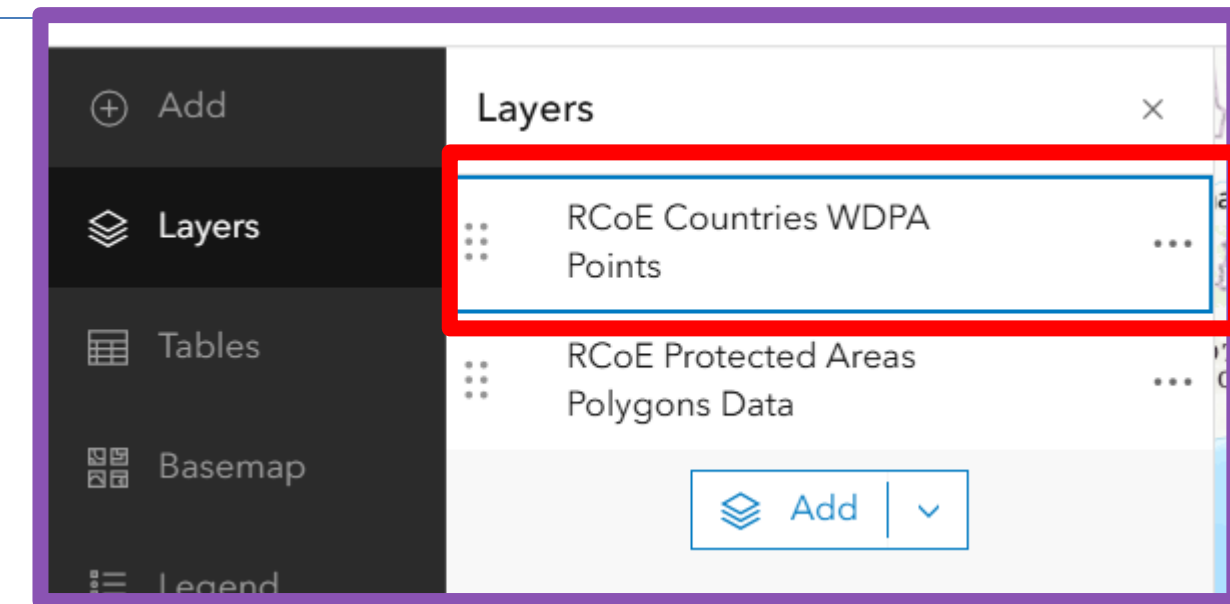
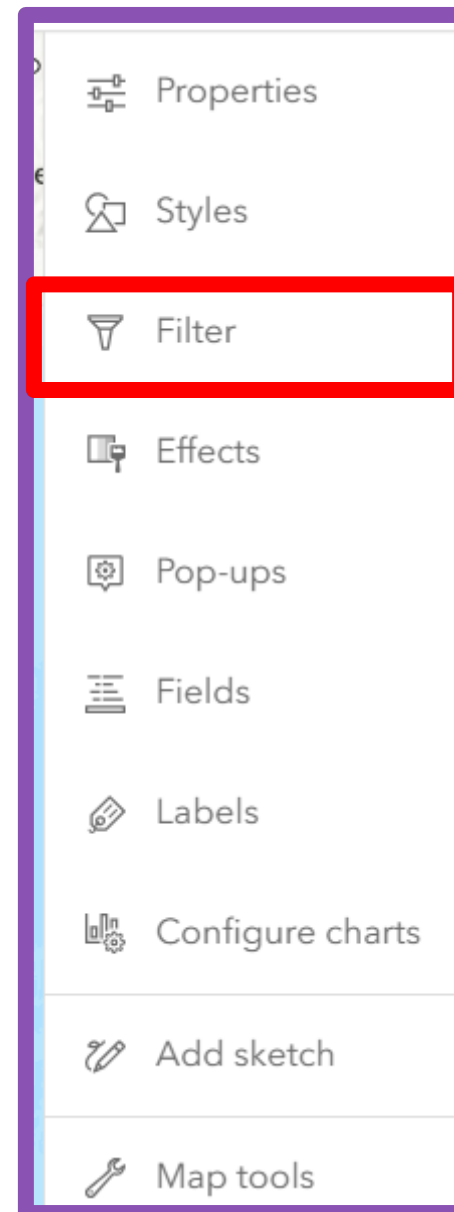
- To save the map click on the save and open option and choose save map.
- Give it a title and select a folder where your map will be saved in.
- Then give it tags – i.e., phrases that can be used to search for your map.
- Lastly write a summary that best describe your map and click on save

A screenshot of a 'Save map' dialog box. The dialog has a white background and a purple border. It contains several fields: 'Title' with the text 'Protected and Conserved Areas of Kenya', 'Folder' with a dropdown menu showing 'Application of GIS in Conservation', 'Tags' with several tags like 'protected areas', 'protected and conserved areas', 'cities', 'world cities', 'airports', and 'world airports', and a 'Summary' text area with the text 'This map shows the protected and conserved areas of Kenya together with cities and international airport.' At the bottom right, there are 'Save' and 'Cancel' buttons. The entire dialog box is highlighted with a red rectangle.

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Filtering data in ArcGIS online

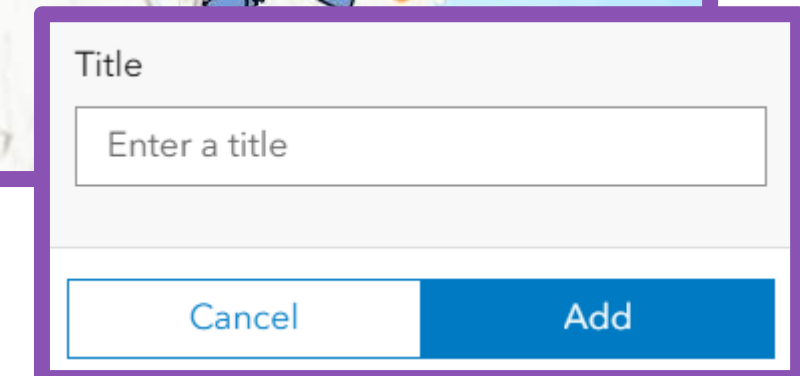
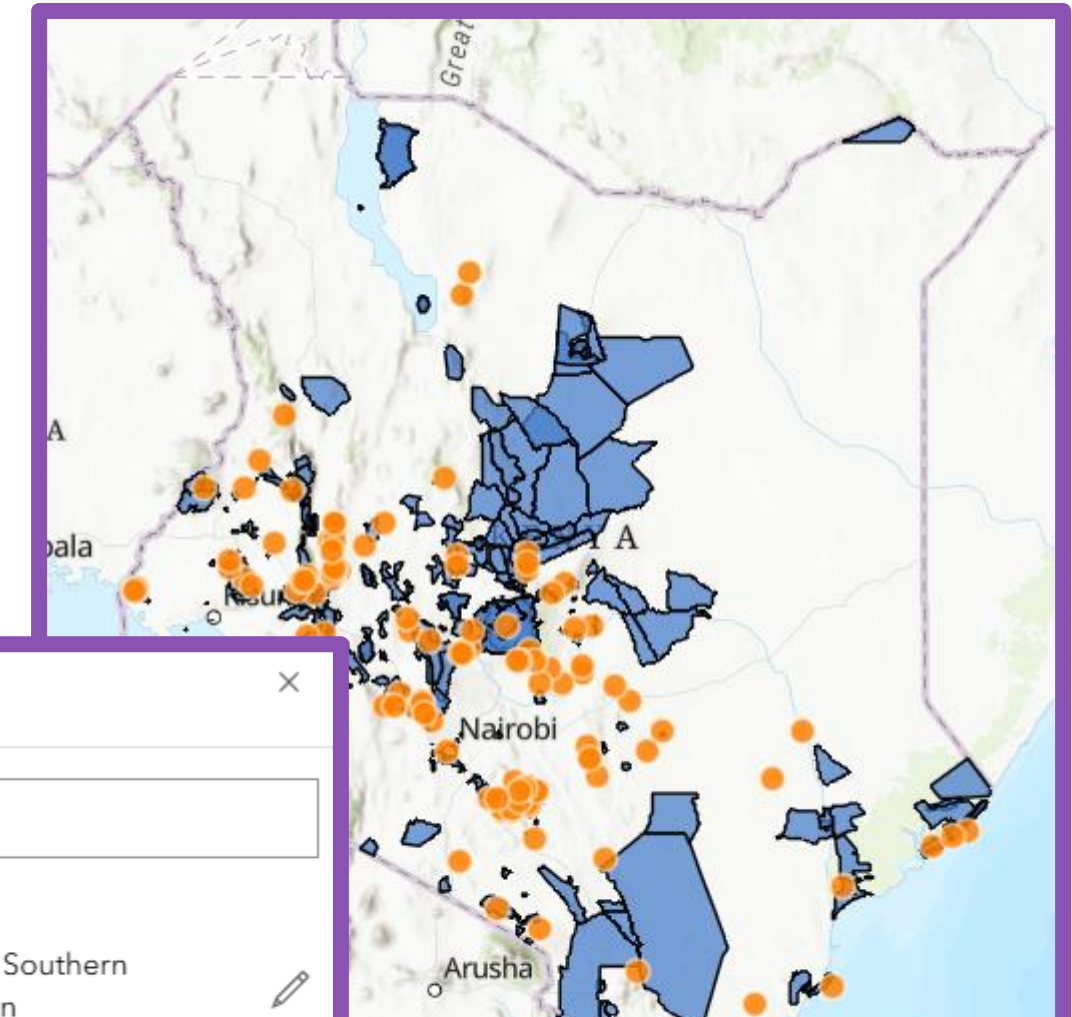
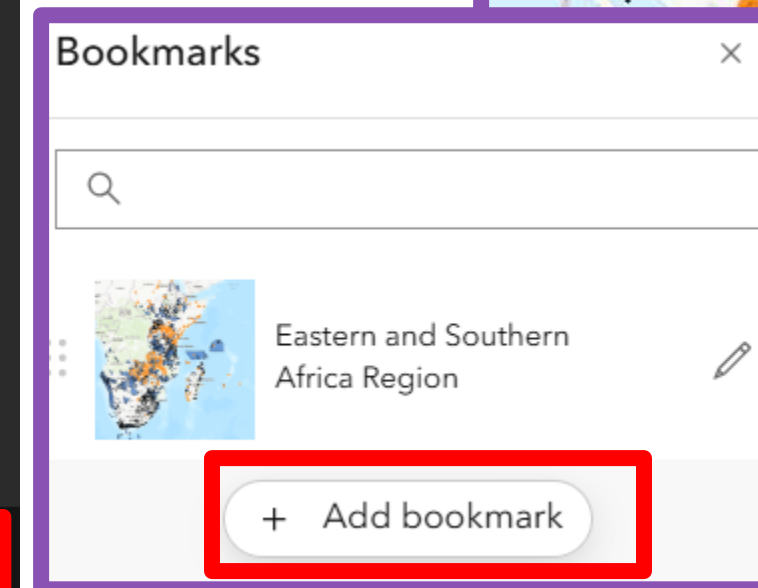
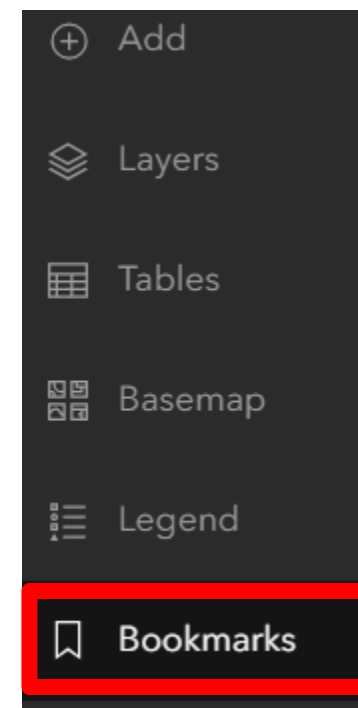
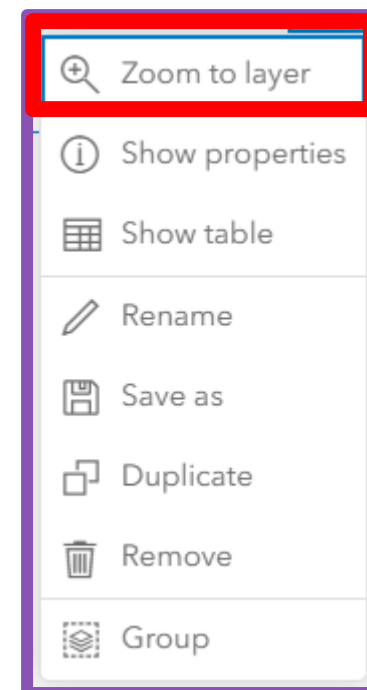
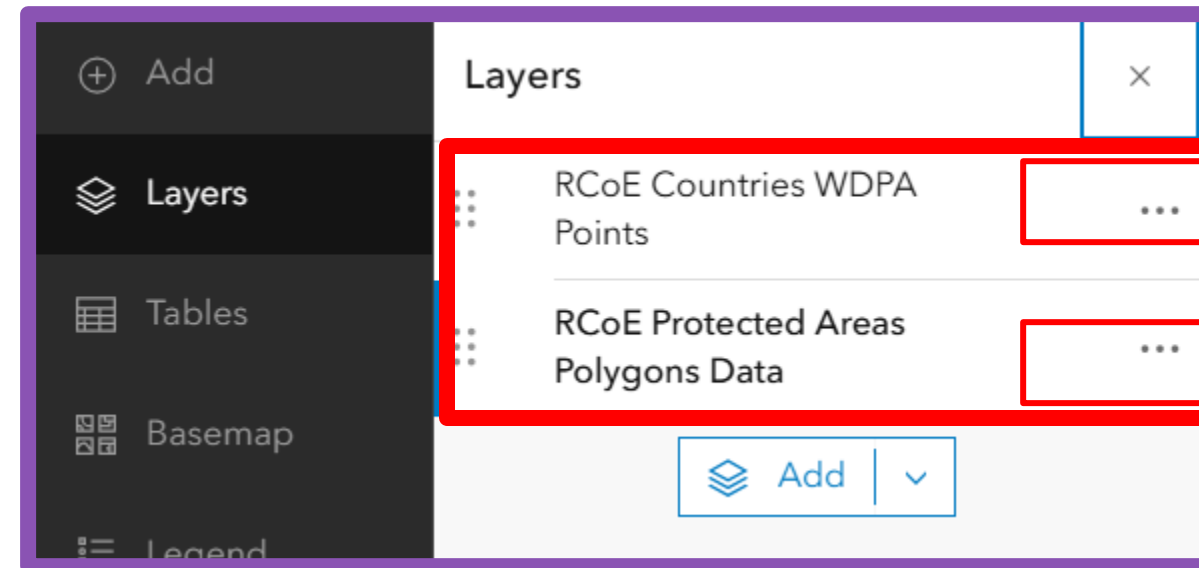
- Let's now learn how to filter out features from a large dataset and only remain with the features of interest.
- Select the layer you want to filter. This time, let's filter the 'RCoE Countries WDPA Points' layer
- On the right side of the window, click on this icon  to expand the widget.
- Click on the 'Filter' option to specify the condition for filtering the features. Filter the data using ISO3 code for your country.
- Click on the save option to apply the filter. Repeat the same procedure to filter the RCoE Protected Areas Polygons Data' layer



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Zoom and Bookmark the view

- If you have several sites of interest in the map, you can bookmark them for easier access in future.
- Now I want us to bookmark the data extent for your country.
- Click on the 3 dots beside the layer name and select 'Zoom to layer' option.
- To bookmark this view, click on the Bookmarks option on the left widget and select the Add bookmark option.
- Enter the title of the bookmark and click on Add option



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Styling features

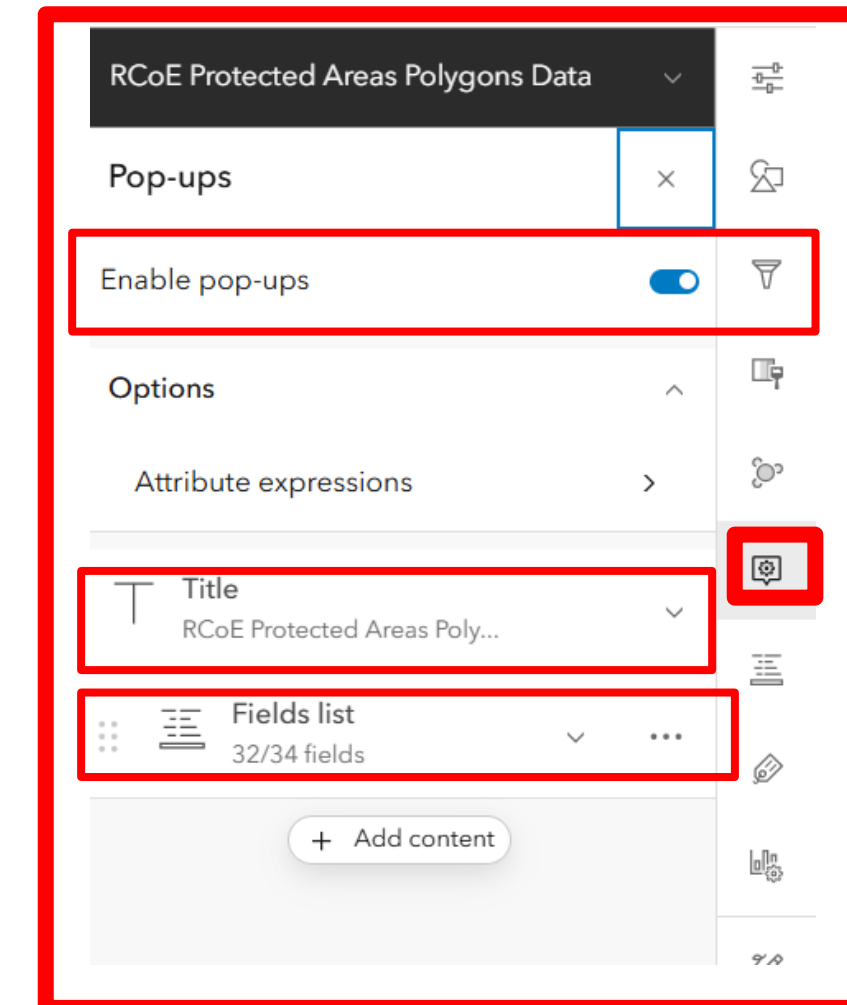
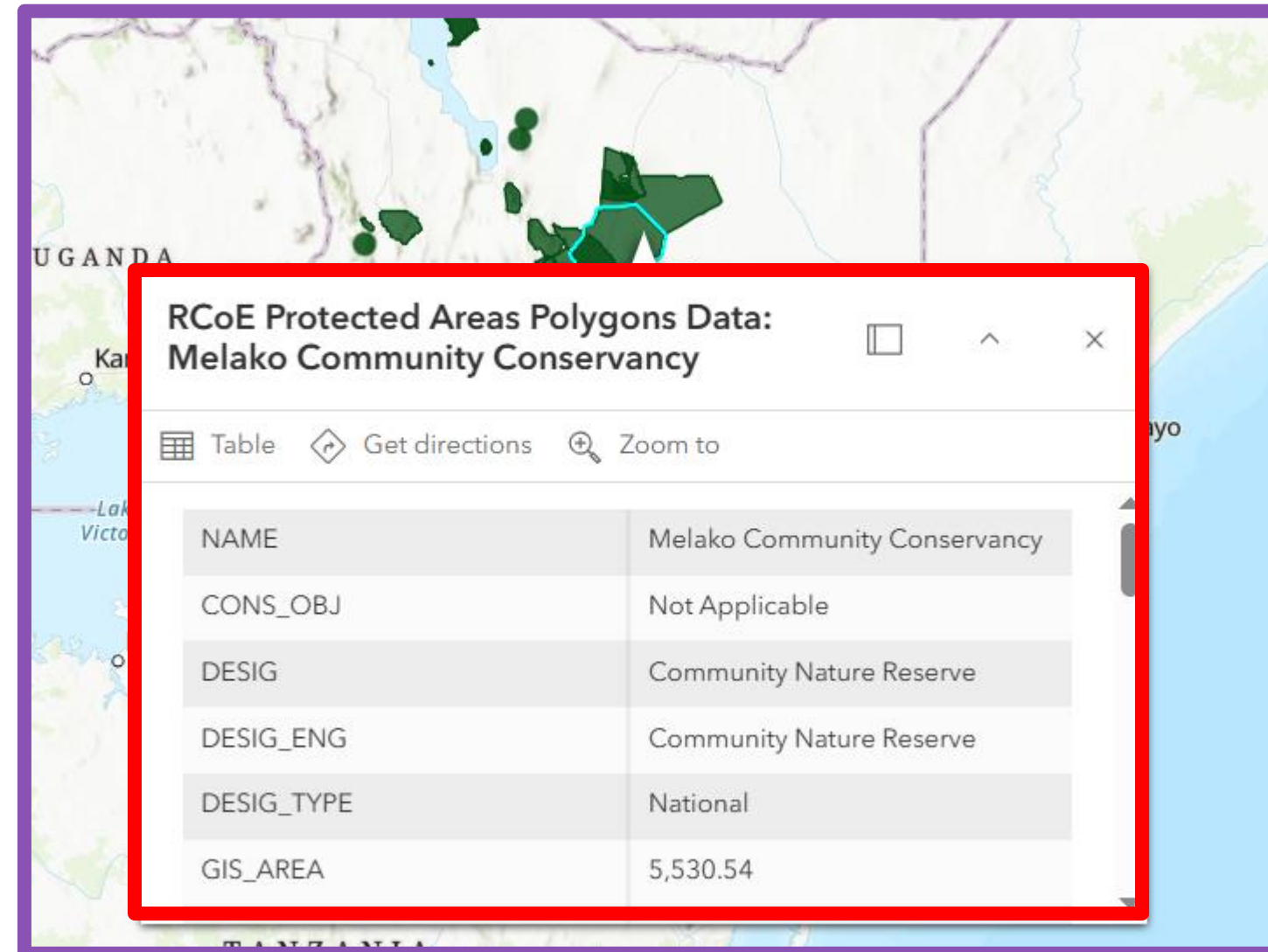
- Change the color of both the points and polygon features to green.
- To change the color, first select the appropriate layer on the left widget under the Layers option.
- Then click on pick a style, the symbol style and change the fill color.
- Select green or enter this hexacode 054d15 then click on done.
- Follow the same procedure to change the color of the other layer
- Save the changes.

The screenshot illustrates the QGIS interface for styling layers. On the left, the 'Layers' panel shows two layers: 'RCoE Countries WDPA Points' and 'RCoE Protected Areas Polygons Data'. The 'Layers' panel is highlighted with a red box. The 'RCoE Countries WDPA Points' layer is selected, and its 'Symbol style' dialog is open. The 'Symbol style' dialog shows the 'Basic point' symbol, and the 'Fill color' is highlighted with a red box. The 'Fill color' is currently orange, and the 'Fill transparency' is set to 15%. The 'Symbol style' dialog is also highlighted with a red box. On the right, the 'Styles' panel is open, showing the 'Choose attributes' and 'Pick a style' sections. The 'Pick a style' section is highlighted with a red box, and the 'Done' button is highlighted with a red box. A map preview is shown in the bottom right corner, displaying the styled data.

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Modifying the Pop-up

- A popup is a window that automatically appears on top of the other window. In ArcGIS online it appears when you click on a feature. It displays the attribute information of a feature.
- Let us modify the popup window to display less and rich(media) information.
- First, select the layer of interest then click on the popup icon on the properties widget located on the right side of the window. You can disable a popup by switching
- You can change the title and use a custom title or select one of the fields in the attribute table to use as the title of the popup.



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Modifying the Pop-up

- You can also change the body text. By the default the popup will display the list of all the fields. You can however remove some of the fields which the users don't need to see.
- You can also display the popup as a description that contains dynamic fields. This is the option we are going to configure.
- Click on the drop down icon beside the 'Fields list' option. Note: you can also specify a title here.
- Enter a description that will be used for the popup. Use this text *'This is {NAME}. It is one of the {DESIG}s in {ISO3}. It covers an area of {GIS_AREA} square kilometers'*. Feel free to use your own text but make sure you include two or more dynamic fields.

The following table represents the data shown in the first screenshot:

Field Name	Value
NAME	Sibilo
CONS_OBJ	Not Applicable
DESIG	National Park

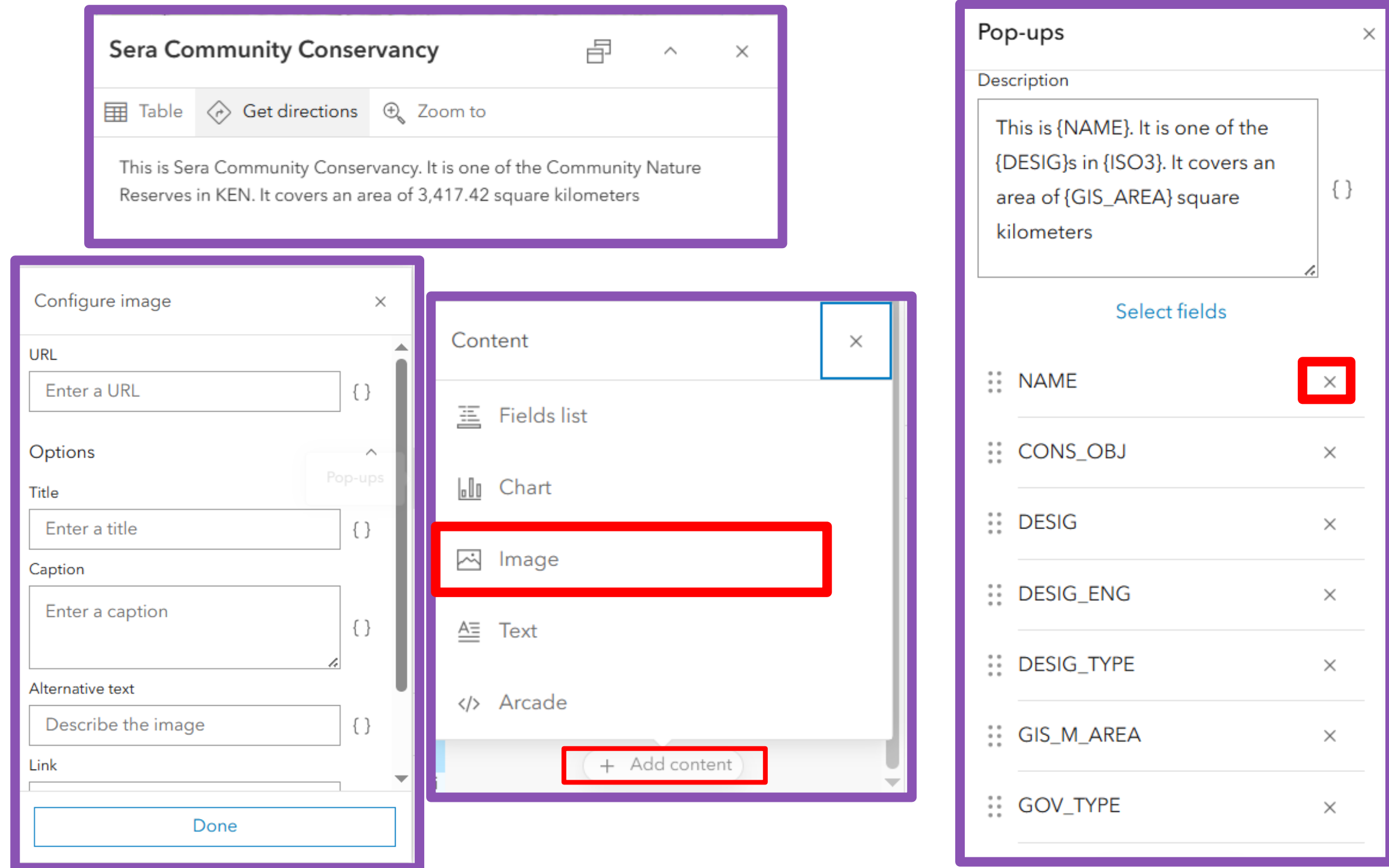
The final screenshot shows the following description text:

This is {NAME}. It is one of the {DESIG}s in {ISO3}. It covers an area of {GIS_AREA} square kilometers

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Modifying the Pop-up

- Now let us delete the list of fields in the default popup. Click on the  beside the field name to remove it from the list. Do it for all the fields. Once all the fields have been removed, your popup should look like the one on the right.
- You can also add an image on the popup. To add an image click on the 'Add content' button and then select image.
- Enter the URL of the image you want to add, the title, the caption, and the link where the users will be redirected when they click on the image.
- Note that adding a URL will add a common image for all the features. To add unique images for each feature you need to use the field option. Create a field and add unique image url for each feature.
- Click on done to effect the changes

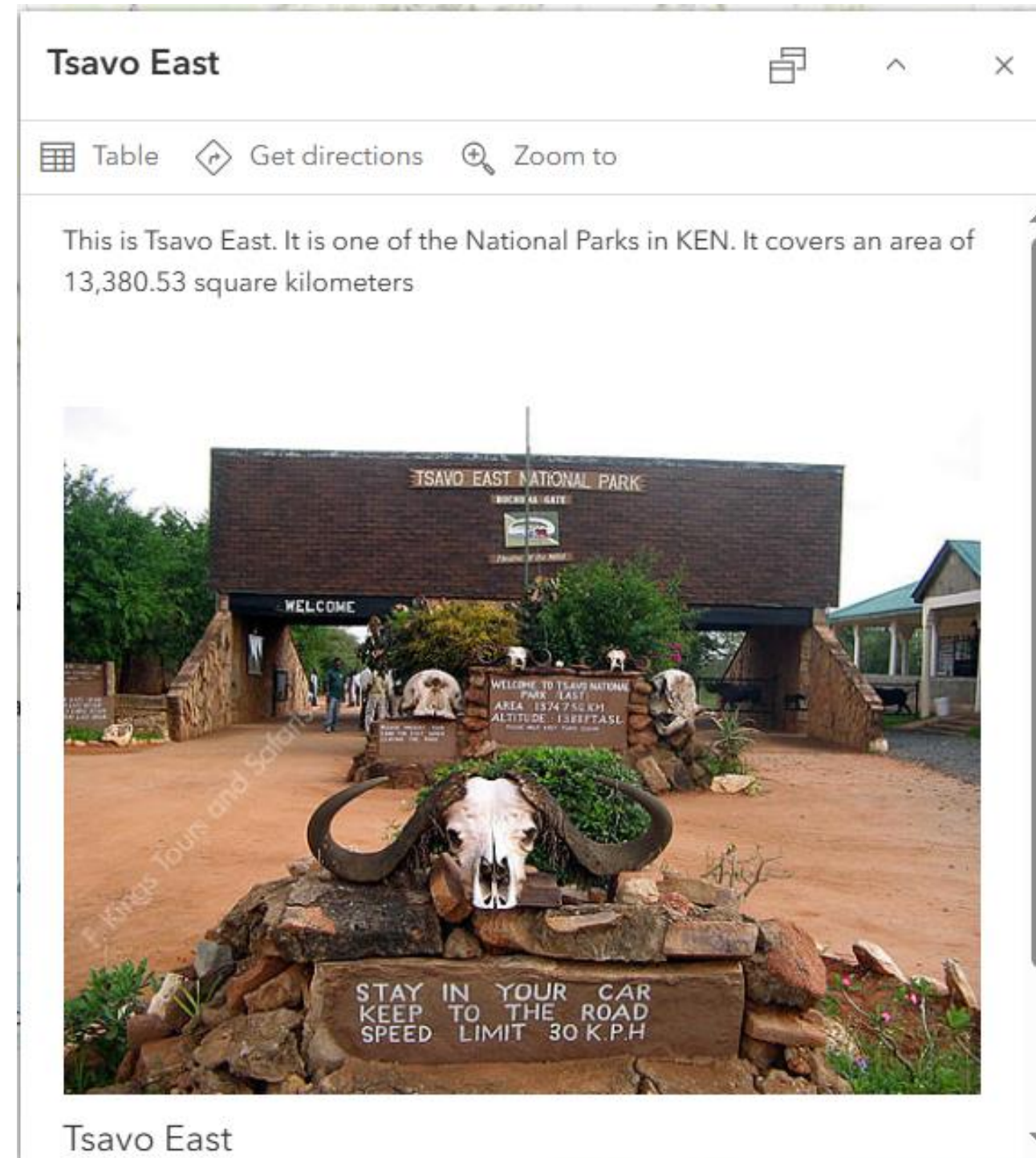


The screenshot illustrates the steps to modify a popup in QGIS. At the top, a popup window for 'Sera Community Conservancy' is shown with a text description. Below it, the 'Configure image' dialog is open, displaying fields for URL, Options, Title, Caption, Alternative text, and Link, along with a 'Done' button. To the right of this dialog is the 'Content' panel, which lists available content types: Fields list, Chart, Image (highlighted with a red box), Text, and Arcade. At the bottom of the Content panel is a red box containing a '+ Add content' button. On the far right, the 'Pop-ups' configuration window is shown, listing various fields (NAME, CONS_OBJ, DESIG, DESIG_ENG, DESIG_TYPE, GIS_M_AREA, GOV_TYPE) with a red box highlighting the 'X' icon next to the 'NAME' field to indicate its removal.

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Modifying the Pop-up

- If you have followed all the procedures, your popup should look like this one.
- **Assignment:** Create a media field in the attribute table of the protected areas of your country and add unique media URLs for all the protected areas in your country. If your country has more than 20 protected areas, select 10 major protected areas for this assignment.



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Sharing your web map

- Ensure that you save the changes you have made to the map by clicking on the save button.
- On the right side of the window, click on the share map and select the 'Manage sharing' option.
- Note that you can share with your colleagues in your organization, or a group of fewer colleagues in your group, or the public.
- Please share your map with 'Everyone (public)' then click on save.
- Lastly, click on share map option, copy the link to your map and submit it as your assignment in the elearning platform.

The screenshot displays the ArcGIS web interface. On the left, a sidebar contains various map tools: Add, Layers, Tables, Basemap, Legend, Bookmarks, Charts, Save and open, Map properties, Share map, Embed map, Information, and Collapse. The 'Share map' option is highlighted with a red box. To its right, the 'Manage sharing' option is also highlighted with a red box. Below these, the 'Copy link to map' option is visible. The main area shows a 'Bookmarks' panel with two entries: 'Eastern and Southern Africa Region' and 'Kenya's Protected and Conserved Areas'. On the right, a 'Share' dialog box is open, showing the 'Set sharing level' section. The 'Owner' option is selected, indicating that 'Owner of the item(s) has access'. Below this, the 'Organization' and 'Everyone (public)' options are listed. The 'Set group sharing' section shows 'None yet' and an 'Edit group sharing' button. At the bottom of the dialog are 'Cancel' and 'Save' buttons. A text box at the bottom of the screenshot contains the URL: <https://arcg.is/0P9nPK0>.

Thank you

Connect with us online

 <https://rcoe-geoportal.rcmrd.org/>

 Regional Centre of Excellence - RCoE

 @RCoE_ESA

 Regional Centre of Excellence (RCoE) for Biodiversity, Forests and Seascape Ecosystems Management

 Regional Centre of Excellence (RCoE) - ESA

